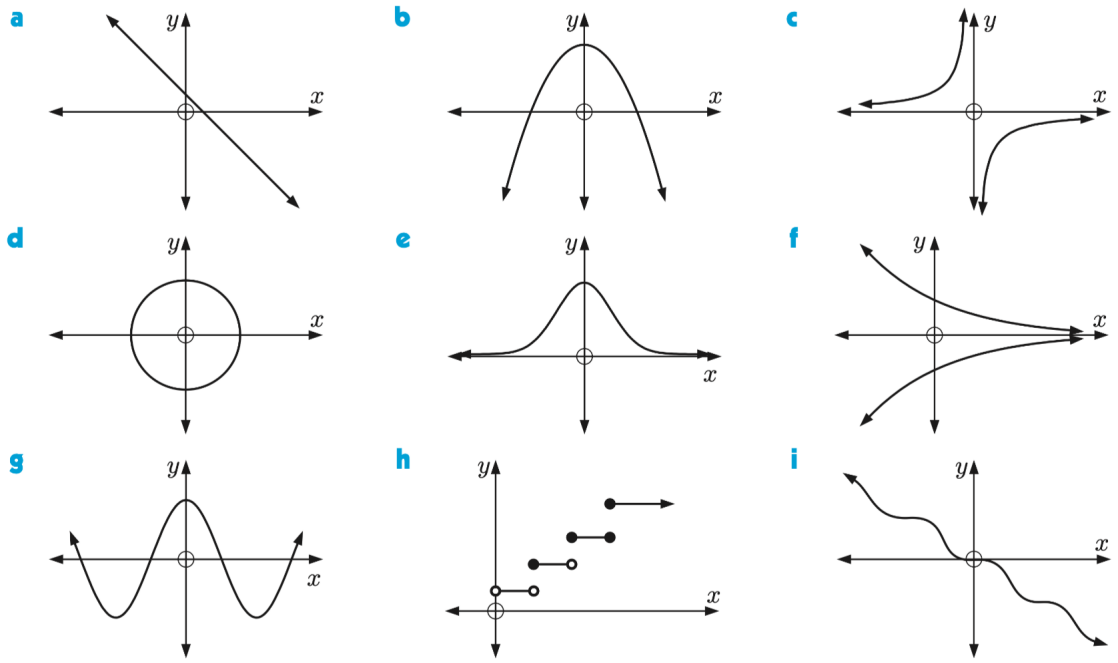
EXERCISE 1 Use the vertical line test to determine which of the following relations are functions :



EXERCISE 2 Give algebraic evidence to show that the relation

$$x^2 + y^2 = 9$$

is not a function.

EXERCISE 3 If $f(x) = 7 - 3x$, find in simplest form :

1. $f(a)$
2. $f(-a)$
3. $f(a+3)$
4. $f(b-1)$
5. $f(x+2)$
6. $f(x+h)$

EXERCISE 4 If $F(x) = 2x^2 + 3x - 1$, find in simplest form :

1. $F(x+4)$
2. $F(2-x)$
3. $F(-x)$
4. $F(x^2)$
5. $F(x^2-1)$
6. $F(x+h)$

EXERCISE 5 Suppose

$$G(x) = \frac{2x+3}{x-4}.$$

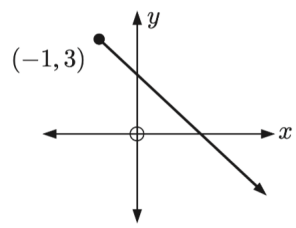
1. Evaluate :
   a. $G(2)$
   b. $G(0)$
   c. $G\left(-\frac{1}{2}\right)$
2. Find a value of x such that $G(x)$ does not exist.
3. Find $G(x+2)$ in simplest form.
4. Find x if $G(x) = -3$.

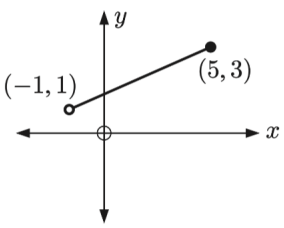
EXERCISE 6 f represents a function. What is the difference in meaning between f and $f(x)$?

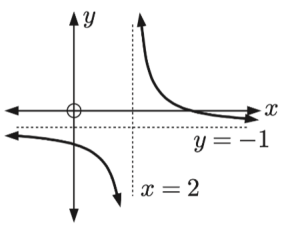
EXERCISE 7

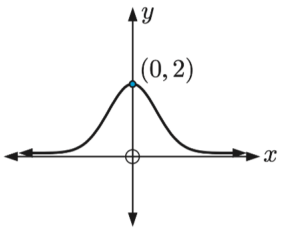
1. Find a linear function $f(x) = ax + b$ for which $f(2) = 1$ and $f(-3) = 11$.
2. Given $f(x) = ax + \frac{b}{x}$, $f(1) = 1$, and $f(2) = 5$, find constants a and b.
3. Given $T(x) = ax^2 + bx + c$, $T(0) = -4$, $T(1) = -2$, and $T(2) = 6$, find a, b, and c.

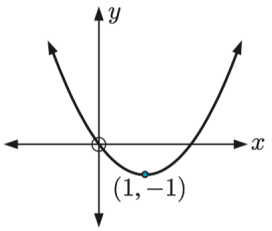
EXERCISE 8 For each of the following graphs, find the domain and range.

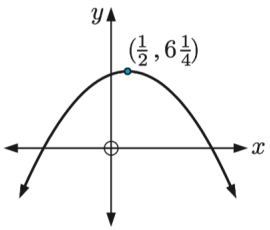
a


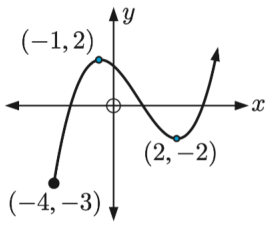
b


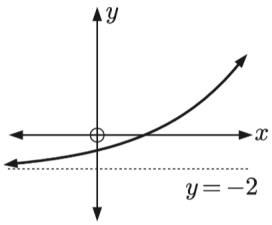
c


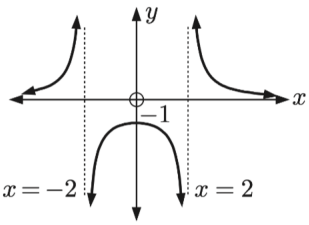
d


e


f


g


h


i


EXERCISE 9 State the values of x for which $f(x)$ is defined, and hence state the domain of the function.

1. $f(x) = \sqrt{x+6}$
2. $f : x \mapsto \dfrac{1}{x^2}$
3. $f(x) = \dfrac{-7}{\sqrt{3-2x}}$

EXERCISE 10 Find the domain and range of each of the following functions :

1. $f : x \mapsto 2x - 1$
2. $f(x) = 3$
3. $f : x \mapsto \sqrt{x}$
4. $f(x) = \dfrac{1}{x+1}$
5. $f(x) = -\dfrac{1}{\sqrt{x}}$
6. $f : x \mapsto \dfrac{1}{3-x}$

EXERCISE 11 Given $f(x) = 2 + x$ and $g(x) = 3 - x$, find :

1. $(f \circ f)(x)$
2. $(f \circ g)(x)$
3. $(g \circ f)(x)$

EXERCISE 12 Given $f(x) = \sqrt{6 - x}$ and $g(x) = 5x - 7$, find :

1. $(g \circ g)(x)$
2. $(f \circ g)(1)$
3. $(g \circ f)(6)$

EXERCISE 13 Given $f : x \mapsto x^2$ and $g : x \mapsto 2 - x$, find $(f \circ g)(x)$ and $(g \circ f)(x)$. Find also the domain and range of $f \circ g$ and $g \circ f$.

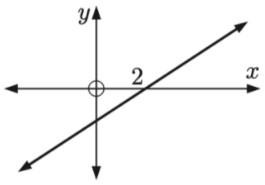
EXERCISE 14 Suppose $f : x \mapsto x^2 + 1$ and $g : x \mapsto 3 - x$.

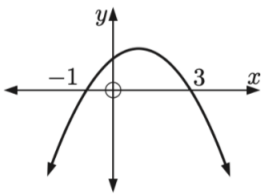
1. Find in simplest form :
   a. $(f \circ g)(x)$
   b. $(g \circ f)(x)$
2. Find the value(s) of x such that $(g \circ f)(x) = f(x)$.

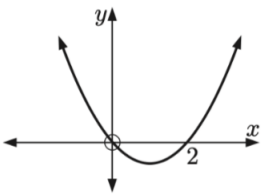
EXERCISE 15 Given $f(x) = \sqrt{1 - x}$ and $g(x) = x^2$, find :

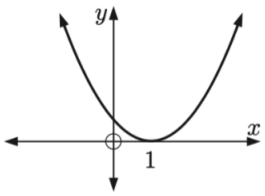
1. $(f \circ g)(x)$
2. the domain and range of $(f \circ g)(x)$

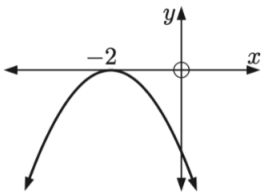
EXERCISE 16 Draw sign diagrams for these graphs :

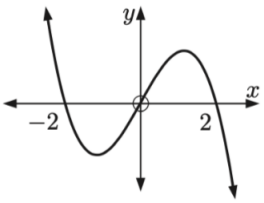
a 

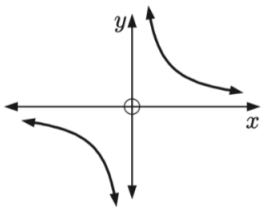
b 

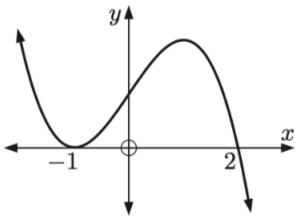
c 

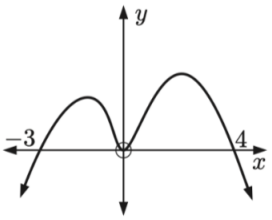
d 

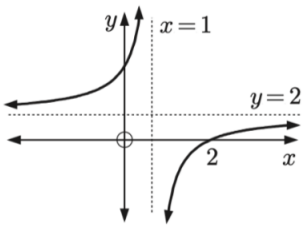
e 

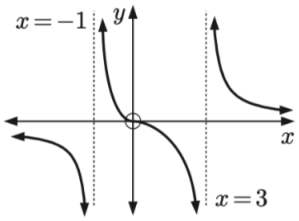
f 

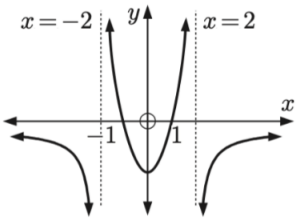
g 

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EXERCISE 17 Draw sign diagrams for :

(a) $(x+4)(x-2)$

(b) $x(x-3)$

(c) $x(x+2)$

(d) $-(x+1)(x-3)$

(e) $(2x-1)(3-x)$

(f) $(5-x)(1-2x)$

(g) x^2-9

(h) $4-x^2$

(i) $5x-x^2$

(j) x^2-3x+2

(k) $2-8x^2$

(l) $6x^2+x-2$

(m) $6-16x-6x^2$

(n) $-2x^2+9x+5$

(o) $-15x^2-x+2$

EXERCISE 18 Draw sign diagrams for :

(a) $\frac{x+2}{x-1}$

(b) $\frac{x}{x+3}$

(c) $\frac{2x+3}{4-x}$

(d) $\frac{4x-1}{2-x}$

(e) $\frac{3x}{x-2}$

(f) $\frac{-8x}{3-x}$

(g) $\frac{(x-1)^2}{x}$

(h) $\frac{4x}{(x+1)^2}$

(i) $\frac{(x+2)(x-1)}{3-x}$

(j) $\frac{x(x-1)}{2-x}$

(k) $\frac{x^2-4}{-x}$

(l) $\frac{3-x}{2x^2-x-6}$

EXERCISE 19 Consider the function

$$y = \frac{ax+b}{cx+d}, \quad \text{where } a, b, c, d \text{ are constants and } c \neq 0.$$

1. State the domain of the function.

2. State the equation of the vertical asymptote.

3. Show that for $c \neq 0$,

$$\frac{ax+b}{cx+d} = \frac{a}{c} + \frac{b-\frac{ad}{c}}{cx+d}.$$

Hence determine the equation of the horizontal asymptote.

EXERCISE 20 For the following functions :

1. Determine the equations of the asymptotes.

2. State the domain and range.

3. Find the axes intercepts.

4. Discuss the behaviour of the function as it approaches its asymptotes.

5. Sketch the graph of the function.

(a) $f : x \mapsto \frac{3}{x-2}$

(b) $y = 2 - \frac{3}{x+1}$

(c) $f : x \mapsto \frac{x+3}{x-2}$

(d) $f(x) = \frac{3x-1}{x+2}$

EXERCISE 21 For each of the following functions f :

(i) On the same set of axes, graph $y = x$, $y = f(x)$, and $y = f^{-1}(x)$.

(ii) Find $f^{-1}(x)$ using coordinate geometry and the gradient of $y = f^{-1}(x)$ from (i).

(iii) Find $f^{-1}(x)$ using variable interchange.

a. $f : x \mapsto 3x + 1$

b. $f : x \mapsto \dfrac{x+2}{4}$

EXERCISE 22 For each of the following functions f :

(i) Find $f^{-1}(x)$.

(ii) Sketch $y = f(x)$, $y = f^{-1}(x)$, and $y = x$ on the same set of axes.

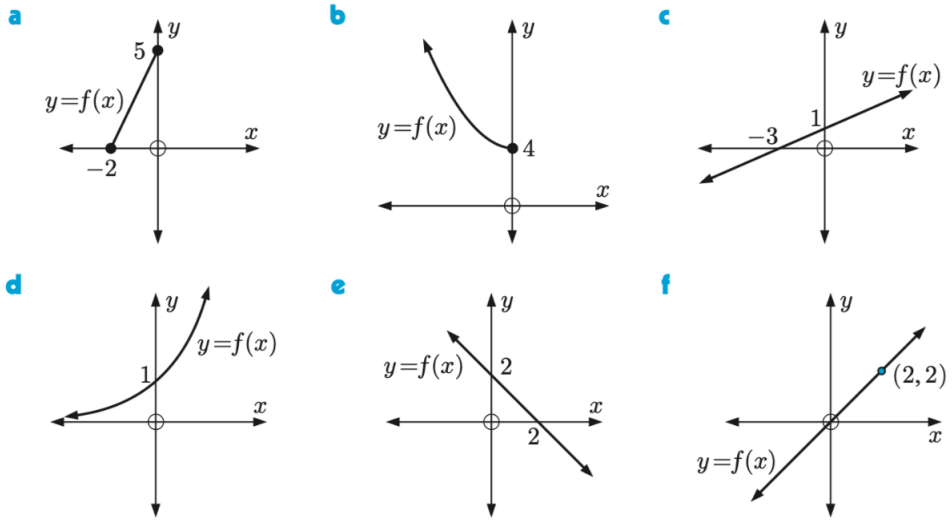
(iii) Show that $(f^{-1} \circ f)(x) = (f \circ f^{-1})(x) = x$, the identity function.

a. $f : x \mapsto 2x + 5$

b. $f : x \mapsto \dfrac{3-2x}{4}$

c. $f : x \mapsto x + 3$

EXERCISE 23 Copy the graphs of the following functions and draw the graphs of $y = x$ and $y = f^{-1}(x)$ on the same set of axes.



EXERCISE 24 Sketch the graph of $f : x \mapsto x^3$ and its inverse function $f^{-1}(x)$.

EXERCISE 25 Given $f : x \mapsto \frac{1}{x}$, $x \neq 0$, find f^{-1} algebraically and show that f is a self-inverse function.

EXERCISE 26 Given $f : x \mapsto 2x$ and $g : x \mapsto 4x - 3$, show that

$$(f^{-1} \circ g^{-1})(x) = (g \circ f)^{-1}(x).$$

EXERCISE 27 Which of these functions is a self-inverse function ?

a. $f(x) = 2x$,

b. $f(x) = x$,

c. $f(x) = -x$,

d. $f(x) = \frac{2}{x}$,

e. $f(x) = \frac{-6}{x}$.